

Fig. 2: Receiver circuit

ostable built around 555 timer (IC4). The high output of the monostable drives the piezobuzzer to sound an audio alarm.

The receiver section operates off 5V DC generated from another 9V battery and voltage regulator IC 7805 (IC2). The battery (Batt.2) is connected to the circuit through switch S2.

When the laser light transmitted through LD1 falls on phototransistor T1, the output of the op-amp (IC3) at its pin 6 remains high. In this condi-

tion, the output of IC4 remains low and the buzzer does not sound.

When the laser beam falling on phototransistor T1 is interrupted by someone, the output of op-amp IC3 goes low and IC3 produces a pulse. This pulse triggers monostable IC4 and its output goes high to sound the alarm for a time period of about $88 \times C8$.

Assemble the transmitter and receiver circuits on separate general-purpose PCBs and enclose in suitable

cabinets. Mount the transmitter and receiver units on opposite pillars of the entrance, aligning the two such that the laser beam from the transmitter directly falls on the phototransistor. Block the laser beam with your hand and measure the op-amp output. It should not be low. At pin 3 of IC4, you should get a positive-going pulse of one-second duration beginning with high-to-low edge of the trigger pulse appearing at pin 2 of IC4 or collector of transistor T2. ●

NANO-AMPERE METER

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This circuit can measure currents down into the nano-ampere range. It is useful when building, testing and experimenting with low-power circuits, especially those operating off batteries.

The circuit is built around commonly available JFET input op-amp TL081 and a few precision resistors. It has a wide operating range from 0-10 nA to 0-1 μ A in steps of multiplicative

decades. The range can be easily extended to 10 μ A and 100 μ A by including additional range resistors of 100

ohms and 10 ohms, respectively.

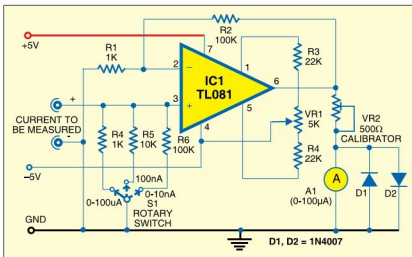


Fig. 1: Circuit of nano-ampere meter

The ± 5 V supply (not shown in the circuit) may be obtained from standard 7805 and 7905 regulators to power the circuit. IC TL081 is wired as a high-gain non-inverting DC amplifier with a gain of $R_f/R_i + 1 = 100$. The current to be measured is passed through a fixed known resistor. The voltage drop across this resistor is applied to the amplifier and output as